

Optimizing Emergency Department Patient Flow & Operational Capacity

Executive summary and Core KPIs

This project analyzes clinical throughput, bottleneck operational friction, and revenue realization patterns across 10,000 Emergency Department (ED) visits. By engineering data-driven metrics like total Door-to-Doctor Wait Time, this analysis isolates critical systemic delays that depress patient satisfaction and cause financial leakage through walkouts. The objective is to convert granular clinical logs into clear operational strategy.

Total ED Patient Volume: 10,000 visits	Hospital Admission Rate: 21.17%
Avg Door-to-Doctor Wait: 78.87 mins	AMA Walkout Rate: 5.41%
Avg Length of Stay (LOS): 3.26 hours	30-Day Readmission Rate: 12.81%
Avg Patient Satisfaction: 6.18 / 10	Total Billed Gross Revenue: \$44.68M USD

Problem statement

Emergency Department overcrowding and prolonged wait times constitute major structural failures in healthcare delivery. When patients face excessive delays, clinical risk increases due to delayed triage. Furthermore, financial leakage via Left Against Medical Advice (AMA) walkouts escalates, causing immediate loss of billable revenue. Finally, operational drag propagates downstream, reducing bed utilization across major clinical departments. This analysis isolates the precise operational thresholds where delay impacts satisfaction, allowing leadership to adjust staffing and capture lost revenue.

Methodology & Feature engineering

Data cleaning, data health audits, and clinical feature engineering were conducted using Python(Pandas, NumPy). Key performance features engineered included: (1) *Total Door-to-Doctor Wait Time* (the sum of triage and doctor rooming queues), (2) *Excessive Wait Threshold* (a binary indicator for doctor wait times exceeding the 120-minute), and (3) *Arrival Queue Intervals* (the duration since the previous clinical arrival to monitor sudden traffic surges).

Throughput & Bottleneck analysis

A. Clinical triage adherence & operational efficiency

Patients are categorized by acuity using the Emergency Severity Index (ESI) from 1 (Resuscitation) to 5 (Non-Urgent). The data confirms that while the clinical triage queue correctly prioritizes high-acuity patients (Acuity 1 assess time is 17.89 min vs. Acuity 5 at 98.21 min), low-acuity cases (Acuity 4 & 5) suffer massive downstream delays, driving satisfaction down to a low of 5.39/10.

Triage Acuity Category	Total Visits	Avg Triage Wait (min)	Avg Doc Wait (min)	Avg Door-Doc wait(min)	Avg LOS (hour)	Admission %	Avg Satisfaction
1- Resuscitation	462	5.45	17.89	23.34	5.84	67.97%	7.08
2 - Emergent	1442	10.40	26.92	37.32	4.94	56.59%	6.87
3 - Urgent	3497	19.11	48.10	67.21	3.62	28.22%	6.40
4 - Less Urgent	3080	22.71	75.21	97.92	2.52	0.00%	5.87
5 - Non-Urgent	1519	25.23	98.21	123.44	1.54	0.00%	5.39

B. Root cause analysis: Lost revenue via AMA walkouts

A critical cluster of 541 patients walked out Against Medical Advice (AMA). The data isolates a vital operational threshold: 16.27% of AMA patients waited longer than 120 minutes before walking out. This represents direct, addressable gross financial leakage.

Patient Disposition	Patient volume	Avg doctor wait (min)	% waited over 2 hours	Avg satisfaction score
Discharged home	6982	68.28	16.99%	6.00
Admitted to hospital	2117	35.65	3.31%	6.70
AMA walkout	541	65.41	16.27%	6.02
Transferred	323	24.55	0.00%	6.89
Expired	37	16.71	0.00%	7.23

C. Department-level efficiency benchmark

General Emergency and Cardiology process the highest caseload volume. However, Trauma demands the highest billing realization average per single encounter (\$5,670.42). Mean doctor wait times are uniformly elevated (~57-61 mins) across all departments, highlighting a systemic institutional throughput constraint rather than isolated staff delays.

Clinical Department	Patient Volume	Avg Doc Wait (min)	Avg LOS (hour)	Avg billed(\$)	Avg satisfaction
Emergency	2792	59.84	3.25	\$3711.31	6.18
Cardiology	1333	61.31	3.20	\$4984.85	6.17
Orthopedics	1229	58.45	3.41	\$4908.20	6.21
Neurology	1007	61.16	3.15	\$4952.32	6.16
Pediatrics	981	57.38	3.26	\$4104.85	6.16
Obs & Gynae	859	59.00	3.20	\$4441.51	6.20
Trauma	830	59.10	3.32	\$5670.42	6.20
General surgery	581	59.94	3.30	\$4810.99	6.13
Psychiatry	388	59.42	3.23	\$3387.70	6.24

Data-driven recommendation

- **Implement a 'Fast-Track' Clinic for Low Acuity Cases:** Acuity tiers 4 and 5 account for over 45% of total volumes but endure the longest throughput delays, severely eroding patient satisfaction. Routing these cases to an attached fast-track clinic staffed by Nurse Practitioners or Physician Assistants will resolve minor cases rapidly and optimize main ED bed utilization.
- **Dynamic Contingency Staffing Realignment:** Cross-referencing hourly arrival volume with the Day-Hour heatmaps shows predictable bottleneck periods during afternoon shifts and specific days of the week. Nursing and physician shift rotations should be restructured dynamically to match these traffic spikes rather than maintaining fixed 8-hour legacy blocks.
- **Introduce an Automated 90-Minute SLA EHR Notification:** Because 16.27% of patients who walk out AMA face delays exceeding two hours, the care coordination team requires an automated EHR alert triggered at the 90-minute wait mark. This enables active, nurse-led intervention to manage or reassure patients before they decide to leave, saving downstream billing loss.